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Abstract

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1 Introduction

Mathias draft: The use of drones in modern warfare has drastically increased in the past few years. In the Russia/Ukraine war, Ukrainian and other Western sources now assign the share of Russian casualties caused by drones from 70-80% (SAB, 2025) to upwards of 90% (Mukhina, 2026). Additionally, drones are increasingly deployed to destroy material targets such as critical infrastructure or military bases, making their presence in war zone airspace even more prevalent (Kirichenko, 2025; Mathewson, 2026) (maybe also cite thorbjørn here).

This increased use has correlated with significant innovation in drone technologies and strategies, thus creating the need for a following surge of development within counter drone technologies (Kirichenko, 2025; Seidaliyeva et al., 2024). Through counter drone technology, military and infrastructure facilities seek to detect, localize and follow foreign drones in the air, possibly with the purpose of eliminating the vessel(s) in question (Samaras et al., 2019). Counter drone technologies and systems usually employ a mixture of acoustic, visual, radio frequency and/or radar sensors to detect drone presence, location, movement, size and/or model (Seidaliyeva et al., 2024). Each sensor type is characterized by certain advantages and disadvantages, meaning that the best detection systems often utilize multiple sensors and technologies in unison to eliminate weak points (Seidaliyeva et al., 2024). However, developing multi-sensor systems also increase complexity and cost for production and maintenance, which can weaken this approach's viability for use cases where mass production or smaller form factors are needed (Samaras et al., 2019).

When evaluating a use case requiring drone detection in a military field operation context, acoustic-based systems possess a number of advantages compared to other methods, including portability, low cost, passive operation, detection of autonomous or wired drones, and independence from lighting conditions (Samaras et al., 2019; Seidaliyeva et al., 2024). Despite these advantages, the current state of available acoustic drone detection technology for field operations is largely limited to larger form factor solutions that require users to carry specialized hardware (Squarehead, 2026). In comparison, portable radio frequency drone detection solutions are widely available with devices at the size of a handheld radio (DroneShield, 2026) or backpack (D-Fend, 2026), however these fail in the detection of wired drones (Seidaliyeva et al., 2024), which have become a large factor in modern battlefields such as the Russia-Ukraine war (Ryan, 2025). This project targets the gap that currently exists for the development of portable acoustic drone detection solutions.

In collaboration with GN and FalCom, this project works with the assumption that the developed
FPS (as of now): **How does a ML-based drone-detection software perform compared to a DSP based model in efficacy as an autonomous support for soldiers on the frontline of Ukraine?**

2 Previous Research

2.1 DSP

Digital signal processing (DSP) provides the foundation for UAV acoustic analysis by transforming raw audio into representations that reveal its temporal, spectral, and spatial structure. However, the role of DSP varies depending on the task. For tasks that require semantic decision-making, such as detection and classification, DSP primarily provides descriptive signal representations (e.g., time-frequency or harmonic features), which learning-based models then use for evaluation (Jekateryńczuk & Piotrowski, 2026).
Several preprocessing steps are common across studies. Amplitude normalization is typically applied so that the peak magnitude of each audio file is scaled to a fixed level. Filtering is also widely used to isolate the frequencies emitted by drones, as most of the acoustic energy produced by drone propellers lies within approximately 200 Hz to 2 kHz. A band-pass filter (BPF) is therefore commonly applied to attenuate environmental noise outside this range. The Butterworth filter is a frequent choice because of its maximally flat passband response and its attenuation of frequencies beyond the cut-off frequencies on both sides. Where background noise is mixed into the signal, noise-reduction techniques (e.g., spectral subtraction) can further suppress it.
The fast Fourier transform (FFT) and short-time Fourier transform (STFT) are then used to generate spectrograms, which allow learning-based models to be trained on a visual time-frequency representation of the analyzed sound. Finally, Mel-frequency cepstral coefficients (MFCCs) are often computed to extract salient features and reduce the dimensionality of the FFT-derived data. (Li et al., 2018; Safavi et al., 2025; Tejera-Berengue et al., 2024; Tongta et al., 2025)

2.2 Machine Learning for Detecting Drones

As UAVs have become more prominent in modern warfare, the need to detect them has grown accordingly. Machine learning (ML) has proven a useful approach to acoustic drone detection, as learned models can recognize the acoustic signature of a drone in difficult conditions. (Al-Emadi et al., 2021; Samaras et al., 2019; Seidaliyeva et al., 2024; Tejera-Berengue et al., 2024). The models used can be divided into classical ML classifiers and deep neural networks.

Classical classifiers, such as support vector machines (SVM), k-nearest neighbours (kNN), random forests and multilayer perceptrons, do not learn features themselves. Instead, they classify compact, hand-crafted features extracted from the audio, most commonly MFCCs (Ahmed et al., 2022; Akbal et al., 2023; Anwar et al., 2019; Jamil et al., 2020). They are fast and need little training data, but their performance depends on how well the chosen features describe the drone sound.

Deep neural networks instead learn features directly from a time-frequency representation of the audio, typically a (Mel-)spectrogram. Convolutional neural networks (CNNs) treat the spectrogram as an image and learn local patterns, such as the harmonic lines produced by the propellers (Al-Emadi et al., 2021; Casabianca & Zhang, 2021; Seo et al., 2018). Deeper variants, such as ResNet, add skip connections and can be combined with attention modules that weight the most informative parts of the input (Liu et al., 2025). Recurrent

neural networks (RNNs), including long short-term memory (LSTM) networks, instead process the audio as a sequence over time (Jeon et al., 2017; Kolamunna et al., 2021). Convolutional recurrent neural networks (CRNNs) combine the two, using convolutional layers to extract features and recurrent layers to model how they change over time (Al-Emadi et al., 2021; Casabianca & Zhang, 2021). More recently, transformer-based models such as the Audio Spectrogram Transformer (AST) have been applied, which use self-attention to relate all parts of the spectrogram to each other (Berg et al., 2025). Finally, several studies combine multiple models or feature types, either by fusing their features or by letting an ensemble of models vote on the final decision (Casabianca & Zhang, 2021; Chatterjee et al., 2024; Dong et al., 2023).
Table 1 summaries the reported performance of a selection of these models.

Table 1: Peer Reviewed Drone Detection Models

Model	**Reported accuracy**	**Paper**
CNN (MFCC)	98.40% (F1 98.32%)	(Al-Emadi et al., 2021)
CRNN	$>$ 90% for drone type ID; detection not quantified, close to CNN	(Al-Emadi et al., 2021)
RNN (LSTM)	$\sim$21 points below CNN	(Al-Emadi et al., 2021)
AUDRON (MFCC + STFT-CNN + RNN + autoencoder fusion)	98.51% binary / 97.11% multiclass	(Chatterjee et al., 2024)
CNN	96.38% binary / 92.94% multiclass	(Chatterjee et al., 2024)
CRNN	94.72% binary / 92.22% multiclass	(Chatterjee et al., 2024)
RNN	75.00% binary / 57.16% multiclass	(Chatterjee et al., 2024)
Soft-voting CNN ensemble (late fusion)	94.73% (91.04% on real-world data)	(Casabianca & Zhang, 2021)
Hard-voting CNN ensemble	94.65% (89.84% on real-world data)	(Chatterjee et al., 2024)
Soft-voting CRNN ensemble	94.12%	(Casabianca & Zhang, 2021)
Hard-voting CRNN ensemble	93.95%	(Casabianca & Zhang, 2021)
Optimized CNN	92.96%	(Casabianca & Zhang, 2021)

Model	Reported accuracy	Paper
Optimized CRNN	92.85%	(Casabianca & Zhang, 2021)
RNN	87.15%	(Casabianca & Zhang, 2021)
Custom CNN (∼5M params)	97.53%	(Berg et al., 2025)
Audio Spectrogram Transformer (AST)	95.71% (99.45% best single run)	(Berg et al., 2025)
ResNet10 + CBAM attention	94.45% (drone type recognition, including heavy-noise conditions)	(Liu et al., 2025)
ResNet18, AlexNet, GoogLeNet, RNN	Lower than ResNet10 + CBAM; exact numbers not given	(Liu et al., 2025)
Two CNNs (Log-Mel + MFCC) fused with evidence theory	94.5% (works up to 50 m)	(Dong et al., 2023)
LSTM (DronePrint)	96% detection (86% with unknown drone types); 90% identification	(Kolamunna et al., 2021)
SVM, Gaussian kernel (MFCC)	98.3% audio only; 98.5% when fused with AlexNet on video	(Jamil et al., 2020)
SVM, polynomial kernel (LPCC)	83.3%	(Jamil et al., 2020)
Fine kNN (TQWT + skinny-pattern features, INCA selection)	99.72% (7 sound classes, 725 samples)	(Akbal et al., 2023)
CNN (STFT spectrograms)	> 98% detection rate, 1.28% false alarms	(Seo et al., 2018)
RNN (best of GMM / CNN / RNN)	F-score 0.80 (not accuracy); GMM and CNN scored lower	(Jeon et al., 2017)
SVM, cubic kernel (MFCC)	∼96.7%	(Anwar et al., 2019)

A common, pitfall recognized in multiple researches is that military drone data is scarce (Al-Emadi et al., 2021; Vilhurin et al., 2026), research done by Al-Emadi et al. (2021) concluded with their studies: data generated by their own AI model was, in combination with their real drone data more than sufficient for training ML models for drone detection. Al-Emadi et al. (2021) confirmed that the same ML model trained on their enhanced dataset performed superior to the same model trained with the original dataset, thereby enabling the possibility generate data is less than desired exists.
AI quote that is interesting: *For your drone project, a claim like "93% recall at 100 m in 45 dB background noise, with 0.2 false alarms per hour" is far more meaningful than "93% accuracy."*

This project focuses on acoustic drone detection because the proposed system relies on microphone signals from a headset rather than visual, Radio frequency (RF) or radar input. Building on a headset also adds little to the soldier's load, since many soldiers already wear one as part of their standard equipment. The most relevant literature is therefore acoustic deep learning, particularly CNNs and hybrid CNN-LSTM to stay robust in noisy conditions (Cai et al., 2025; Ding et al., 2024; Muhaciroğlu & Akgül, 2026). Surveys note that most results are experimental and hard to compare, partly because standardized reference datasets are lacking (Sliti & Garai, 2023).

Acoustic detection is an active research field within UAV sensing because drone sound can be captured passively and supports detection when visibility is poor or line of sight is blocked (Jarocky et al., 2021; Reuter et al., 2026; Tejera-Berengue et al., 2024). Early systems on FFT-based representations (Fast Fourier Transform) and simple classifiers shows that: one real-time system reached 83% accuracy by matching plotted FFT images against reference templates, and 61% with k-nearest neighbors (Kim et al., 2017). Deep learning has since raised performance substantially CNN, ResNet18, CRNN and EfficientNet-B0 models trained on Mel-spectrograms of open source drone recordings all reached 94-95% accuracy on test data from the same sources (Muhaciroğlu & Akgül, 2026), and fine-tuning a pretrained CNN on a custom acoustic dataset gave a mean average precision of 0.88 on unseen recordings (Yaacoub et al., 2022).

Performance depends heavily on how well the training data match real conditions. Generalizing to unseen recording setups, environments and drone types is a central challenge in acoustic drone detection (Reuter et al., 2026). Al-Emadi et al. (2021) found that models trained on recorded drone audio degraded noticeably on drone types they had not seen during training, and that enlarging the dataset with drone sounds generated by a generative adversarial network (GAN) clearly improved detection of those unseen drones. Muhaciroğlu and Akgül (2026) report the same pattern with field data. CNN, ResNet-18, CRNN and EfficientNet-B0 models trained only on open-source datasets fell to 55-58% accuracy when tested on field recordings of three drone types captured outdoors at 5 m to 2 k from the drone, sampled at 44.1 kHz. Adding those field recordings to the training data raised accuracy to 93-98%, with EfficentNet-B0 reaching 98.05%. A model pretrained on the general-purpose AudioSet detected more drones in out-of-domain data than the same model trained from scratch (true positive rate 0.78 versus 0.70 at a 1% false-positive rate), and augmentations such as noise mixing and microphone simulation gave further gains (Reuter et al., 2026). Distance also matters, in a study using microphone-array recordings mixed with ambient noise, detectors trained on unattenuated signals worked up to 200 m, while training on attenuated signals extended the range to about 300 m for most methods and 500 m for YAMNet-based detector (Tejera-Berengue et al., 2024). The same study found that training on signals with a lower signal-to-noise ratio (SNR) improved overall detector performance. Noisy, realistic conditions should therefore be represented in both training and evaluation.

Among detection models, a clear trend is the move from plain CNNs toward hybrid“¡ architectures. A CNN-LSTM model using MFCC features reached 99.50% detection accuracy and 97.52% recognition accuracy, although its input came from a fiber-optic acoustic sensor rather than a conventional microphone (Ding et al., 2024). Transformer components are also gaining ground. A conformer model combining CNN and Transformer layers, fed

with MFCC and GFCC features, improved accuracy and noise robustness over conventional techniques (Cai et al., 2025).

Localization is a related but separate problem. Microphone-array methods can estimate a drone's direction of arrival (DoA). TSNetIQ, for example, combines Transformer and CNN modules to estimate DoA from raw I/Q signals and remains robust across SNRs, smapling rates and array configurations (Zhu et al., 2025). A recent review notes that UAV monitoring pairs lightweight models with spatiotemporal Transformers to cope with non-stationary noise and edge-computing limits, while the trade-off between accuracy and real-time performance remains a core challenge (Xu et al., 2025). This project asks only whether a drone is present in headset microphone audio, so localization work serves as background rather than as a methodological basis.

2.3 Implementing Drone Detection in a Military Field Context

In military field scenarios, certain conditions apply that are relevant for the development of an acoustic drone detection system. Through research and corroborating interviews with experts and professionals in the Danish defense and military technology industry, some of the conditions were uncovered.

In the modern battlefield, fast-flying offensive drones can make it functionally impossible for soldiers to counterattack the drones by shooting them down (Købsted, 2026, ll.1̃92–199). The sheer speed of drone attacks on military field personnel is also a main cause for the deadliness of drone warfare , as it is often even impossible for soldiers to duck into cover quickly enough (Verner, 2026, ll.1̃46–169). Therefore,

3 Method

3.1 What do we want to figure out?

How does an experimental ML-based system perform in a drone detection simulation compared to a control DSP-based system in the different environmental parameters; wind speed, drone speed and war-themed noises?

Design criteria

- **Criterion 1:** Detect drones 3 seconds before impact
- **Criterion 2:** Use a FalCom Pro Pods microphone
- **Criterion 3:** Simulate Ukrainen war environment

3.2 How do we want to figure it out?

Testing facility and simulation procedure
The experimental product and the control product will be tested under the same circumstances. For the iterative process a virtual simulation test will be conducted, and for the final test the simulations will be played using a surround sound speaker array in an anechoic room.

One testing round consist of 2 constant parameters and 4 dynamic parameters shown in the table below. All dynamic variables will be tested independently of each other. Therefore, we can find the complete number of test rounds $n = W*L*O*V+D = 101.100$ of 20 seconds WAV files with 10 second drone sound in each, where D is dummy clips with no drones sound (1 per each wind and noise level = 100).

Table 2: Parameters of the simulations

Static parameters	Variable	Value
Full wav length	F	20 seconds
Drone playback length	P	10 seconds
Dynamic parameters	Variable	Possibilities
Wind Speed	W	10
Noise Level	L	10
Drone speed	V	101
Drone playback offset	O	10

**Note: Drone playback offset is found by O = F - P*

Example:
Test Round 1: W = 0, L = 0, V = 100, O = 0
Test Round 2: W = 0, L = 0, V = 100, O = 1
...
Test Round 10: W = 0, L = 0, V = 100, O = 9
Test Round 11: W = 0, L = 0, V = 101, O = 0
...
Test Round 1001: W = 0, L = 0, V = 200, O = 0
Test Round 1001: W = 0, L = 0, V = 200, O = 1

W, L and V can be treated as independent parameters, and each can be isolated and evaluated with Confidence over time, where Confidence is calculated from the percentage of drone hops during a window. This can be visualized in a graph of Confidence over time, where a threshold of 3 seconds determines success and the flow of the graph introduces a discussion of the products performance during the specific parameters. Essential: How confident do we detect drones during the different parameters.

3.3 Methodology 2.0

Drone klip bliver lige så lang som test round fordi vi har war shaped noise i stedet for. Så derfor bliver alle detection også true positives, og det er kun vores dummys der bliver false positives.
10 wind, 20 war noise levels, 100 drone speed, 1 offset. 20.000 test rounds

3.4 Iterativ process dagbog

ITERATION 0
0.1
Både DSP og CNN detektere droner.
DSP detekterer droner mens, der ikke er nogen drone. Det gør CNN ikke.
CNN detekterer bedre de hurtige droner end langsomme.

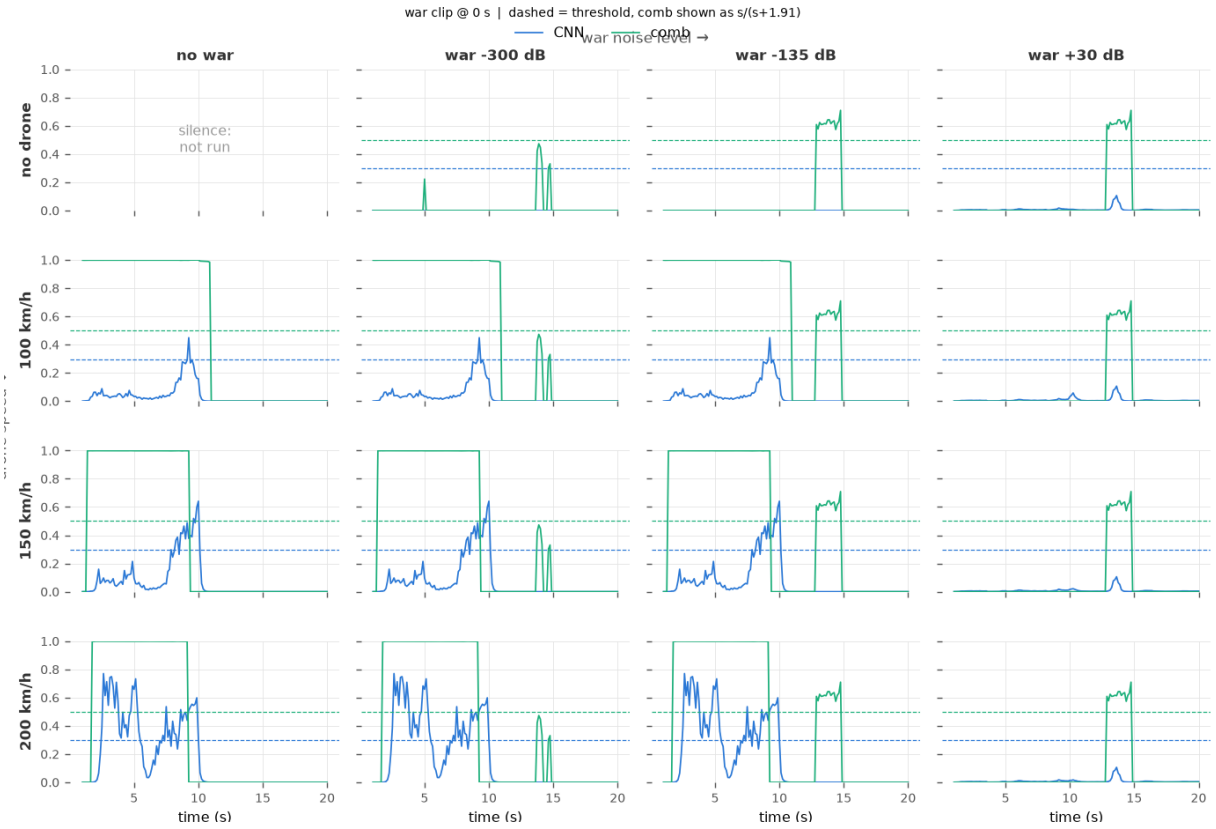


Figure 1: Virtual Iteration 1

0.2
Efter rettelser på test formatet ser vi nu både lydstyrke over tid, frekvens og tid, og confidence over tid. Ved at samligne ser at vi detektere omkirng 250 frekvens. Det kan være derfor at 200 km/t bliver deteketeret best. Dette kan også være frekvens blandet med sound level. Men det ser ikke ud til at være sound level udelukkende
Vi skal detektere lavere frekvenser bedre!!!!
Måske skal vi opnå dette ved at træne på vores egen drone?
FEJL FUNDET: VORES DRONE LYD ER IKKE REN
0.3
Okay vores lyd er ren, men vi kan bare ikke høre det. Og vi detekter meget bedre rigtige droner end syntetisk. Til gengæld detekter vi også rigtig meget støj. War shaped noise bliver nemt detekteret uden noget baggrunds støj
1.1
WarShapedNoise ser ud til at være et problem. Vi ved ikke om det repræsentere War ordentligt, men vi opfanger ingen droner igennem det. Vi overvejer at gå tilbage til War Sound Scapes i stedet:
Billedet viser hvordan vi fint kan detektere droner gennem war sound scapes. Vi kan igen konkludere at når droner accelere, bliver det svært at detektere dem. Det kan skyldes den store mængde stille stående droner i test sampled. Mads siger at claude siger at faktisk har testede accelerende droner, men mine svar viser at det er der jeg ikke kan detekte dem. Hmm. Lyver han?
1.2
Iteration 6 ser ud til at være meget bedre end iteration 9. Det ser ud til at være fordi vi tester på en drone fra trænings settet.

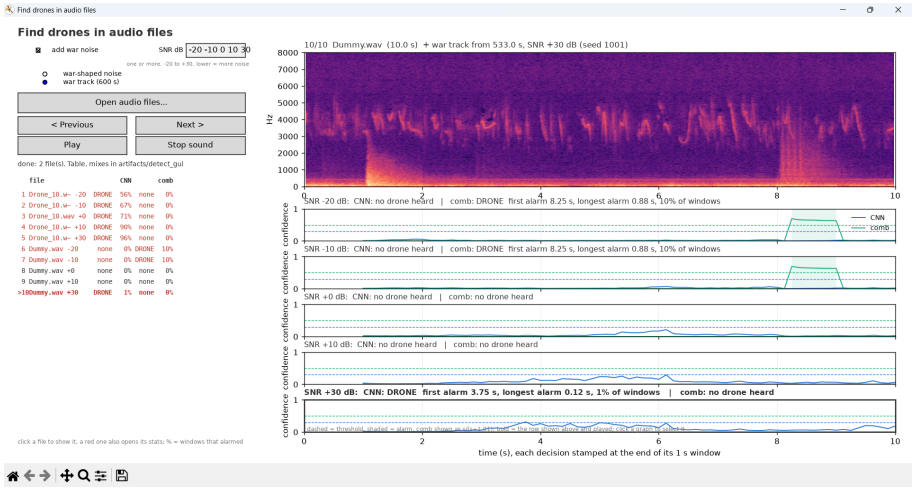


Figure 2: Enter Caption

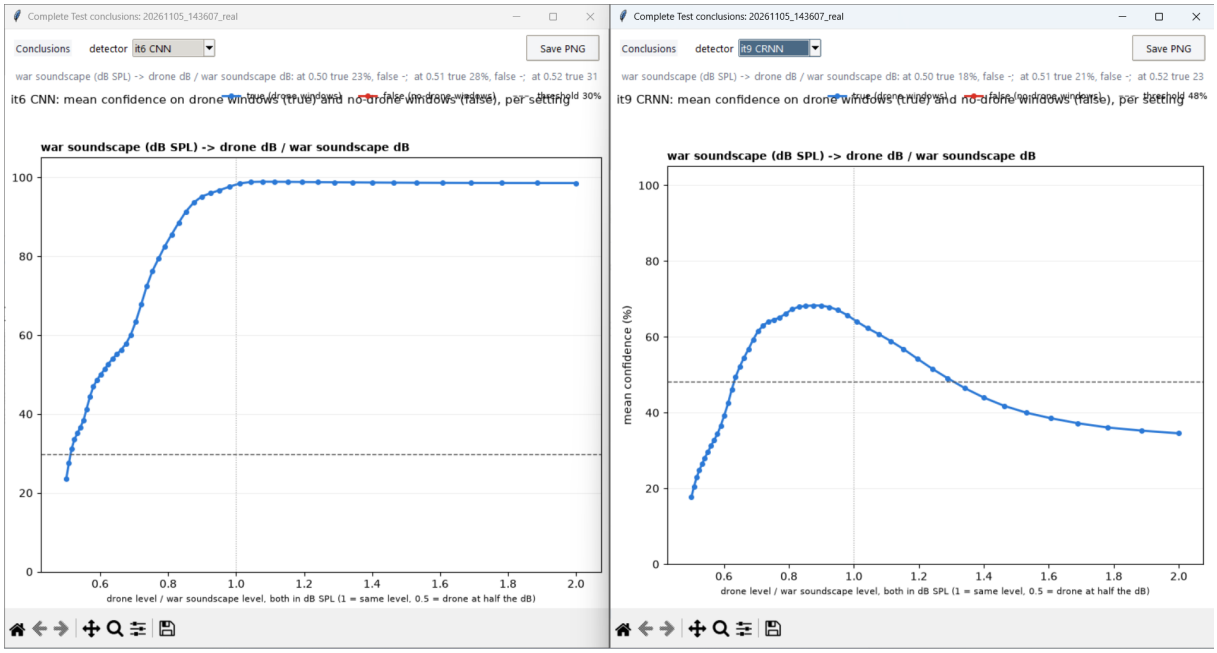


Figure 3: Enter Caption

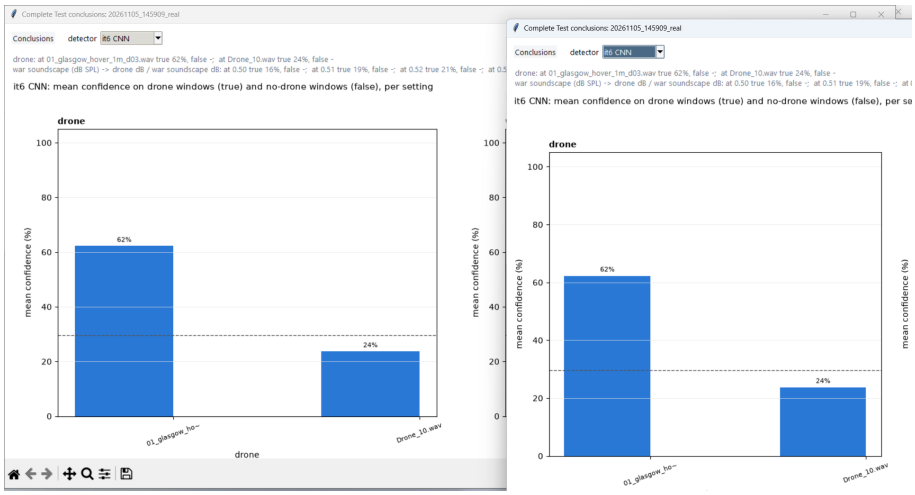


Figure 4: Enter Caption

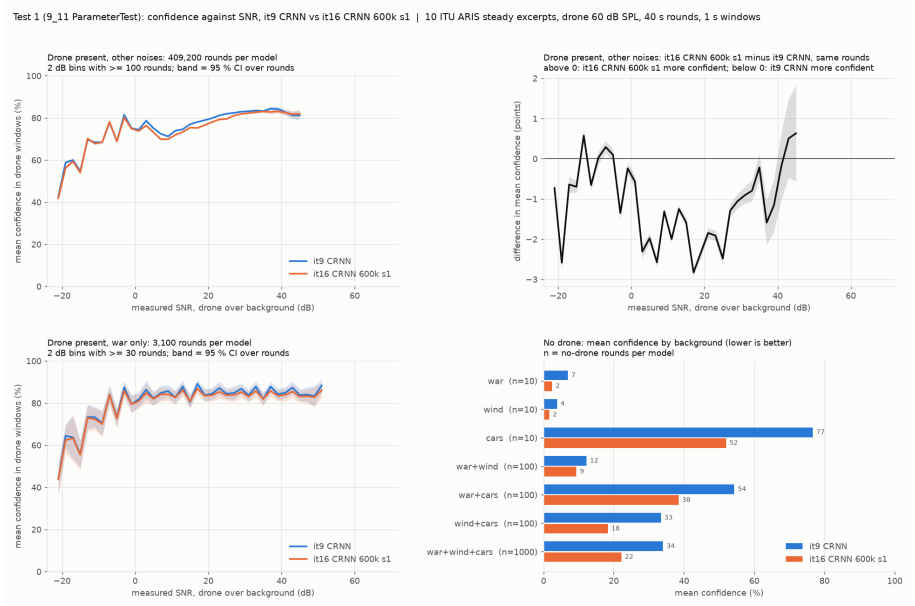


Figure 5: Enter Caption

Iteration 6 er klart den mest sikre. Men det er den kun på droner den har trænet på
2.0 Kærer dagbog. Ved godt det er lang tid siden. Vi har lige kørt den største test nogensinde: 700.000 test rounds. Update: Vi vil lave 3 test
1) Real Drone: Parameter Test
2) Modified Drone: TBI Test
3)
It 16 is better for no false alarms but it9 is better for detecting drones. Right false alaram is a bigger problem so it 16 might be the focus.
2.1 Iteration colors is working really well.
2.2
Nu er vi der hvor vi bare tester forskellige iterationer. Dette er eksempel hvor vi tester iteration forskellige s 1'ere. Konklussionen er IT19 Color er den bedste
The difference between the confidence of a drone, and the confidence of a drone in distractors is key. Because that makes it easier to do a threshold between the detection of

Model	1. Med drone, krigslyd	2. Uden drone, distraktorer
it9 CRNN	**81,4 %**	33,9 % (dårligst)
it16 CRNN 600k s1	80,7 %	**22,2 %** (bedst)

Figure 6: Enter Caption

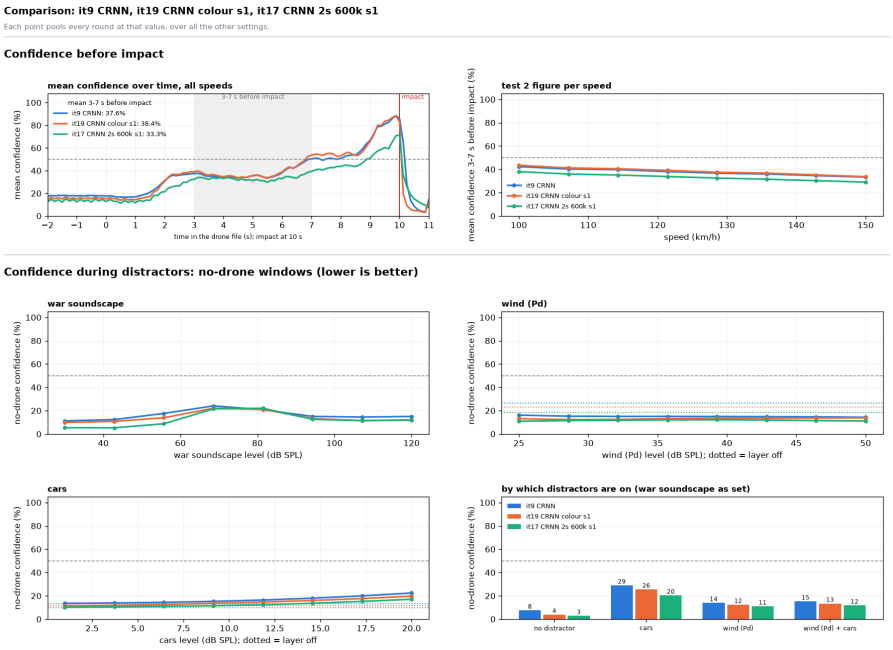


Figure 7: Enter Caption

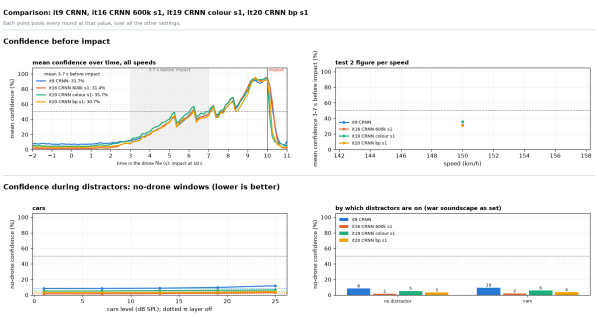


Figure 8: w

drones and the detection of distractors.
3.0
Vi har lige fået drone fra Dan. Det først vi gør er at teste hvor stor indflydelse det har på vores confidence.
3.1
Vi har lige implementeret en buffer. Det betyder at lige efter (standard er et sekundet efter) der er har været impact fra en drone. Så tæller confidence ikke som false positive.

4 Design & Implementation

5 Findings

6 Discussion

Test-set overfitting is when we change and iterate our model to better score on our test set. If done incorrectly, it will result in a model performing well on the exact test set. However it will worsen its efficacy on actual drone in actual environments.

7 Future Work

8 Conclusion

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Appendix